Code No: 132AA **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech I Year II Semester Examinations, July/August - 2021
ENGINEERING PHYSICS – II
(Common to EEE, ECE, CSE, EIE, IT, ETM)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Explain the concept of dual nature of light.
b) Describe the experimental verification of matter waves using Davisson Germer experiment? [7+8]

2. State and explain Heisenberg's uncertainty principle. Using uncertainity principle explain the absence of electrons in the nucleus. [15]

3.a) Derive an expression for carrier concentration in p-type semiconductor.
b) Explain the formation of PN junction. And draw the I-V characteristics of PN junction. [7+8]

4. Derive an expression for carrier concentration in intrinsic semiconductors. [15]

5. Explain electronic polarizability in atoms and obtain an expression for electronic polarizability in terms of radius of the atom. [15]

6.a) Derive an expression for calculation of internal field for a cubic dielectric crystal.
b) Explain Piezoelectricity, pyroelectricity and ferroelectricity and give examples of each. [7+8]

7.a) Explain classification of dia, para and ferro magnetic materials on the basis of magnetic moment.
b) How are superconductors classified? Explain Meissner effect. Write application of superconductivity. [7+8]

8. What are nanomaterials? Explain the three methods for the fabrication of nanomaterials. [15]

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